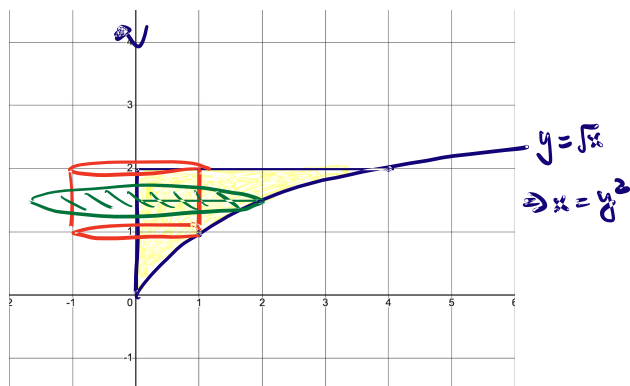




1. Consider the region bounded by $y = \sqrt{x}$, y -axis and $y = 2$.

1). Sketch the graph of the above region, and make sure to shade the region.



Cylindrical shell! " dx "

Disk " dy "

2). Set up integrals to compute the volume of the solid of revolution obtained by rotating the region in Part 1) about the y -axis using both disk/washer and cylindrical shell methods.

i). Disk/Washer Method:

$$\text{Volume of the Solid} = \int_0^2 \pi (y^2)^2 dy$$



ii). Cylindrical Shell Method:

$$\text{Volume of the Solid} = \int_0^4 2\pi x (2 - \sqrt{x}) dx$$

3). Use either one of the integrals in Part 2) to compute the volume of the solid obtained by rotating the given region about y -axis. Simplify the answer.

$$\begin{aligned} V &= \int_0^2 \pi y^4 dy = \pi \cdot \frac{1}{5} y^5 \Big|_0^2 \\ &= \frac{32\pi}{5} \end{aligned}$$